39

FOR IBPS SBI PO/CLERK PRELIMS 2026

QUANT CHECKLIST

the PRACTICE PAPER

/STUDIFIEDPDF


COMPLETE
QUANT EXCLUSIVE
12 Months 24 Months
₹3375 ₹4305
₹1451 ₹1851
USE CODE TT57
OFFER VALID TILL
17 th FEBRUARY
BY AASHISH ARORA

Dear Students,

This daily practice sheet is designed to build both speed and accuracy, one day at a time.

It contains a mix of easy, moderate, and challenging questions to prepare you for every possible scenario in the exam. Treat it like a warm-up before the real game.

Solve it daily without fail. Don't wait for motivation—show up with discipline. Because it's not talent but consistent hard work that takes you places.

Stay focused. Stay consistent. Work hard.

-Aashish Arora

CONTENTS

1. SIMPLIFICATION AND APPROXIMATION

Direction: What value should come in place of the question mark (?) in the following question?

1. 68.75% of 320 + $\frac{1}{8}$ of 256 - √324 = ?

A. 230

B. 232

C. 234

D. 238

E. None of these

2. [$\frac{11}{16}$ of 288 + $\frac{5}{12}$ of 240] ÷ 4 + 3^2 = ?

A. 83.5

B. 81.5

C. 85.5

D. 87.5

E. None of these

3. 2 $\frac{2}{5}$ of 75 - 14.28% of 350 + 13 = ?

A. 137

B. 139

C. 141

D. 143

E. None of these

4. $\sqrt[3]{5832}$ + $\frac{7}{12}$ of 192 - 31.25% of 128 = ?

A. 84

B. 86

C. 88

D. 92

E. None of these

5. [$\frac{3}{8}$ of 640 + √1600] ÷ 5 + $\frac{1}{4}$ of 84 = ?

A. 73

B. 77

C. 79

D. 81

E. None of these

6. 44.44% of 180 + $\frac{2}{11}$ of 242 + 5^2 = ?

A. 149

B. 145

C. 147

D. 151

E. None of these

7. (17^2 - 13^2) ÷ 8 + 62.5% of 96 = ?

A. 69

B. 72

C. 78

D. 75

E. None of these

8. $1\frac{3}{7}$ of 98 + 22.22% of 180 - √196 = ?

A. 158

B. 160

C. 162

D. 168

E. None of these

9. [$\frac{13}{40}$ of 400 + $\frac{5}{16}$ of 256] ÷ 5 + 2^3 = ?

A. 46

B. 50

C. 52

D. 54

E. None of these

10. 83.33% of 72 + 11.11% of 216 + $\frac{1}{6}$ of 90 = ?

A. 95

B. 97

C. 99

D. 101

E. None of these

11. [√2304 ÷ 4] + $\frac{14}{15}$ of 150 - $\frac{1}{5}$ of 85 = ?

A. 129

B. 132

C. 138

D. 135

E. None of these

12. $\frac{5}{12}$ of 372 - 6.25% of 320 + $\sqrt[3]{1728}$ = ?

A. 141

B. 143

C. 145

D. 149

E. None of these

13. [$\frac{9}{16}$ of 512 - $\frac{3}{16}$ of 256] ÷ 4 + 7^2 = ?

A. 109

B. 105

C. 111

D. 113

E. None of these

14. $2\frac{5}{8}$ of 64 + 31.25% of 160 - $\frac{2}{5}$ of 95 = ?

A. 170

B. 175

C. 180

D. 185

E. None of these

15. [$\frac{4}{7}$ of 350 + √576] ÷ 7 + 12.5% of 96 = ?

A. 40

B. 44

C. 46

D. 48

E. None of these

16. $\frac{7}{9}$ of 324 + $\frac{13}{40}$ of 160 - $\frac{3}{8}$ of 128 = ?

A. 248

B. 252

C. 258

D. 260

E. None of these

17. 1 $\frac{1}{4}$ of 96 + 53.33% of 150 - √625 = ?

A. 165

B. 170

C. 180

D. 175

E. None of these

18. (15^2 - 9^2) ÷ 6 + $\frac{8}{15}$ of 225 - 14.28% of 350 = ?

A. 94

B. 90

C. 92

D. 96

E. None of these

19. [68.75% of 640 ÷ 16] + [$\frac{11}{16}$ of 176] - 2^2 = ?

A. 140.5

B. 142.5

C. 146.5

D. 148.5

E. None of these

20. 1 $\frac{5}{6}$ of 84 + 18.75% of 256 + $\frac{1}{13}$ of 520 = ?

A. 236

B. 238

C. 240

D. 244

E. None of these

Answers:

1. C

2. A

3. D

4. E
5. B
6. A
7. D
8. E
9. B
10. C
11. D
12. E
13. A
14. C
15. B
16. E
17. D
18. A
19. E
20. E

Solutions:

1. 68.75% of 320 + $\frac{1}{8}$ of 256 - √324

= 220 + 32 - 18

= 252 - 18

= 234

2. [$\frac{11}{16}$ of 288 + $\frac{5}{12}$ of 240] ÷ 4 + 3^2

= [198 + 100] ÷ 4 + 9

= 298 ÷ 4 + 9

= 74.5 + 9

= 83.5

3. 2 $\frac{2}{5}$ of 75 - 14.28% of 350 + 13

= 180 - 50 + 13

= 130 + 13

= 143

4. $\sqrt[3]{5832}$ + $\frac{7}{12}$ of 192 - 31.25% of 128

= 18 + 112 - 40

= 130 - 40

= 90

5. [$\frac{3}{8}$ of 640 + √1600] ÷ 5 + $\frac{1}{4}$ of 84

= [240 + 40] ÷ 5 + 21

= 280 ÷ 5 + 21

= 56 + 21

= 77

6. 44.44% of 180 + $\frac{2}{11}$ of 242 + 5^2

= 80 + 44 + 25

= 124 + 25

= 149

7. (17^2 - 13^2) ÷ 8 + 62.5% of 96

= (289 - 169) ÷ 8 + 60

= 120 ÷ 8 + 60

= 15 + 60

= 75

8. $1\frac{3}{7}$ of 98 + 22.22% of 180 - √196

= 140 + 40 - 14

= 180 - 14

= 166

9. [$\frac{13}{40}$ of 400 + $\frac{5}{16}$ of 256] ÷ 5 + 2^3

= [130 + 80] ÷ 5 + 8

= 210 ÷ 5 + 8

= 42 + 8

= 50

10. 83.33% of 72 + 11.11% of 216 + $\frac{1}{6}$ of 90

= 60 + 24 + 15

= 84 + 15

= 99

11. [√2304 ÷ 4] + $\frac{14}{15}$ of 150 - $\frac{1}{5}$ of 85

= [48 ÷ 4] + 140 - 17

= 12 + 140 - 17

= 152 - 17

= 135

12. $\frac{5}{12}$ of 372 - 6.25% of 320 + $\sqrt[3]{1728}$

= 155 - 20 + 12

= 135 + 12

= 147

13. [$\frac{9}{16}$ of 512 - $\frac{3}{16}$ of 256] ÷ 4 + 7^2

= [288 - 48] ÷ 4 + 49

= 240 ÷ 4 + 49

= 60 + 49

= 109

14. $2\frac{5}{8}$ of 64 + 31.25% of 160 - $\frac{2}{5}$ of 95

= 168 + 50 - 38

= 218 - 38

= 180

15. [$\frac{4}{7}$ of 350 + √576] ÷ 7 + 12.5% of 96

= [200 + 24] ÷ 7 + 12

= 224 ÷ 7 + 12

= 32 + 12

= 44

16. $\frac{7}{9}$ of 324 + $\frac{13}{40}$ of 160 - $\frac{3}{8}$ of 128

= 252 + 52 - 48

= 304 - 48

= 256

17. $1\frac{1}{4}$ of 96 + 53.33% of 150 - √625

= 120 + 80 - 25

= 200 - 25

= 175

18. $(15^2 - 9^2) \div 6 + \frac{8}{15}$ of 225 - 14.28% of 350

= (225 - 81) ÷ 6 + 120 - 50

= 144 ÷ 6 + 120 - 50

= 24 + 120 - 50

= 94

19. [68.75% of 640 ÷ 16] + [$\frac{11}{16}$ of 176] - 2^2

= [440 ÷ 16] + 121 - 4

= 27.5 + 121 - 4

= 148.5 - 4

= 144.5

20. $1\frac{5}{6}$ of 84 + 18.75% of 256 + $\frac{1}{13}$ of 520

= 154 + 48 + 40

= 202 + 40

= 242

2. ARITHMETIC QUESTIONS

Q1: The marked prices of two articles A and B are in the ratio 7:9 respectively. A discount of 20% is allowed on article A, and its selling price becomes Rs. 560. Find the marked price of article B.
दो वस्तुओं A और B के अंकित मूल्य क्रमशः 7:9 के अनुपात में हैं। वस्तु A पर 20% की छूट दी जाती है और उसका विक्रय मूल्य Rs. 560 हो जाता है। वस्तु B का अंकित मूल्य ज्ञात कीजिए।
(A) 840
(B) 900
(C) 960
(D) 880
(E) 980

Q2: The speeds of trains P and Q are in the ratio 5:7. Their lengths are 220 m and 308 m respectively. Running in the same direction, train Q completely crosses train P in 44 seconds. Find the speed of train Q in m/s.
रेल P और Q की गतियाँ 5:7 के अनुपात में हैं। उनकी लंबाइयाँ क्रमशः 220 मीटर और 308 मीटर हैं। समान दिशा में चलते हुए, रेल Q, रेल P को 44 सेकंड में पूरी तरह पार कर जाती है। रेल Q की गति m/s में ज्ञात कीजिए।
(A) 36
(B) 48
(C) 40
(D) 42
(E) 45

Q3: X, Y and Z together can complete a piece of work in 6 days. Y alone can complete the same work in 24 days. Z is 50% more efficient than Y. In how many days can X and Z together complete the work?
X, Y और Z मिलकर एक कार्य को 6 दिनों में पूरा कर सकते हैं। Y अकेला उसी कार्य को 24 दिनों में पूरा कर सकता है। Z, Y से 50% अधिक कुशल है। X और Z मिलकर उस कार्य को कितने दिनों में पूरा करेंगे?
(A) 6
(B) 7

(C) 8
(D) 9
(E) 10

Q4: A tank contains 90 litres of a mixture of milk and water in the ratio 7:3 respectively. If 15 litres of milk and 9 litres of water are added to the tank, find the quantity of milk in the new mixture.
एक टंकी में 90 लीटर दूध और पानी का मिश्रण 7:3 के अनुपात में है। यदि इसमें 15 लीटर दूध और 9 लीटर पानी और मिला दिया जाए, तो नए मिश्रण में दूध की मात्रा ज्ञात कीजिए।
(A) 72
(B) 75
(C) 80
(D) 76
(E) 78

Q5: The average weight of five boys Anuj, Bharat, Chetan, Deepak and Eshan is 44 kg. The average weight of Anuj and Bharat is 42 kg, and the average weight of Deepak and Eshan is 50 kg. Find the weight of Chetan.
पाँच लड़कों अनुज, भरत, चेतन, दीपक और ईशान का औसत वजन 44 किग्रा है। अनुज और भरत का औसत वजन 42 किग्रा है तथा दीपक और ईशान का औसत वजन 50 किग्रा है। चेतन का वजन ज्ञात कीजिए।
(A) 36
(B) 34
(C) 38
(D) 40
(E) 32

Q6: Deepak invested Rs x in a partnership, while Sameer invested Rs 900 more than Deepak for the same period of time. If Deepak's share out of a total profit of Rs 9900 was Rs 4500, find the value of x.
दीपक ने एक साझेदारी में x रुपये निवेश किए, जबकि समीर ने दीपक से 900 रुपये अधिक उसी समयावधि के लिए निवेश किए। यदि 9900 रुपये के कुल लाभ में से दीपक का हिस्सा 4500 रुपये था, तो x का मान ज्ञात कीजिए।
(A) 4200
(B) 4800
(C) 5000
(D) 5400

(E) 4500

Q7: Eight years ago, the sum of the ages of Harsh and Tarun was 24 years. The ratio of their present ages is 5:3. Find Tarun's present age.
आठ वर्ष पहले, हर्ष और तरुण की आयु का योग 24 वर्ष था। उनकी वर्तमान आयु का अनुपात 5:3 है। तरुण की वर्तमान आयु ज्ञात कीजिए।
(A) 12
(B) 15
(C) 18
(D) 20
(E) 24

Q8: Megha spends 25% of her monthly income on household needs, 20% of the remaining amount on coaching classes, and 30% of the amount then left on transport. If she saves Rs 16800, find her monthly income.
मेघा अपनी मासिक आय का 25% घरेलू आवश्यकताओं पर, शेष राशि का 20% कोचिंग कक्षाओं पर, और उसके बाद बची राशि का 30% परिवहन पर खर्च करती है। यदि वह 16800 रुपये बचाती है, तो उसकी मासिक आय ज्ञात कीजिए।
(A) 38000
(B) 42000
(C) 36000
(D) 40000
(E) 44000

Q9: Sneha invested Rs 12000 in Scheme A at 15% simple interest for 4 years. She then invested only the interest earned from Scheme A in Scheme B at 25% compound interest for 2 years. Find the compound interest received from Scheme B.
स्नेहा ने योजना A में 12000 रुपये 15% वार्षिक साधारण ब्याज पर 4 वर्षों के लिए निवेश किए। फिर उसने योजना A से प्राप्त केवल ब्याज राशि को योजना B में 25% वार्षिक चक्रवृद्धि ब्याज पर 2 वर्षों के लिए निवेश किया। योजना B से प्राप्त चक्रवृद्धि ब्याज ज्ञात कीजिए।
(A) 3900
(B) 4200
(C) 4500
(D) 4320
(E) 4050

Q10: The sum of two positive integers A and B is 29. The square of A is 61 more than three times the square of B. Find the value of A - B.
दो धनात्मक पूर्णांकों A और B का योग 29 है। A का वर्ग, B के वर्ग के तीन गुने से 61 अधिक है। A - B का मान ज्ञात कीजिए।
(A) 7
(B) 8
(C) 9
(D) 10
(E) 11

Q11: A badminton racket was sold for Rs 1500. If it had been sold for Rs 180 more, the profit percentage would have been 15 percentage points higher. Find the cost price of the racket.
एक बैडमिंटन रैकेट Rs 1500 में बेचा गया। यदि इसे Rs 180 अधिक में बेचा जाता, तो लाभ प्रतिशत 15 प्रतिशत अंक अधिक होता। रैकेट का क्रय मूल्य ज्ञात कीजिए।
(A) Rs 1100
(B) Rs 1200
(C) Rs 1250
(D) Rs 1350
(E) Rs 1400

Q12: The ratio of the height of a cone to its radius is 4:3. If the volume of the cone is 324π cubic cm, find its slant height.
एक शंकु की ऊँचाई और त्रिज्या का अनुपात 4:3 है। यदि शंकु का आयतन 324π घन सेमी है, तो उसकी तिर्यक ऊँचाई ज्ञात कीजिए।
(A) 12 cm
(B) 14 cm
(C) 16 cm
(D) 15 cm
(E) 18 cm

Q13: The population of a city in 2020 was X. It decreased by 25% in 2021 and then increased by 20% in 2022. If the population in 2022 was 7200, find the population in 2021.
किसी शहर की जनसंख्या वर्ष 2020 में X थी। वर्ष 2021 में इसमें 25% की कमी हुई और फिर वर्ष 2022 में 20% की वृद्धि हुई। यदि वर्ष 2022 में जनसंख्या 7200 थी, तो वर्ष 2021 की जनसंख्या ज्ञात कीजिए।

(A) 5400
(B) 5600
(C) 6000
(D) 6400
(E) 6600

Q14: Rohit and Varun together invested Rs 7800 in a business. The ratio of the time periods for which Rohit and Varun invested is 3:2. If the total profit is Rs 5000 and Rohit's share is Rs 2000, find Varun's investment.
रोहित और वरुण ने मिलकर एक व्यवसाय में Rs 7800 निवेश किए। रोहित और वरुण के निवेश की समय-अवधि का अनुपात 3:2 है। यदि कुल लाभ Rs 5000 है और रोहित का लाभांश Rs 2000 है, तो वरुण का निवेश ज्ञात कीजिए।
(A) Rs 3600
(B) Rs 4200
(C) Rs 4800
(D) Rs 5200
(E) Rs 5400

Q15: The average of 9 numbers is 46. The average of the first 4 numbers is 39, and the average of the last 4 numbers is 50. Find the 5th number.
9 संख्याओं का औसत 46 है। पहली 4 संख्याओं का औसत 39 है और अंतिम 4 संख्याओं का औसत 50 है। 5वीं संख्या ज्ञात कीजिए।
(A) 58
(B) 56
(C) 60
(D) 54
(E) 62

Q16: Rohan and Suresh started a business with investments of Rs 6000 and Rs 9000 respectively. After 5 months, Rohan withdrew all his money and Nitin joined the business with an amount equal to 80% of the total initial investment of Rohan and Suresh. If the total profit at the end of the year was Rs 2960, find Nitin's share of the profit.
रोहन और सुरेश ने क्रमशः 6000 रुपये और 9000 रुपये लगाकर एक व्यवसाय शुरू किया। 5 महीने बाद, रोहन ने अपनी पूरी राशि निकाल ली और नितिन व्यवसाय में रोहन और सुरेश के कुल प्रारंभिक निवेश के 80% के बराबर राशि के साथ शामिल हुआ। यदि वर्ष के अंत में कुल लाभ 2960 रुपये था, तो नितिन का लाभांश ज्ञात कीजिए।

(A) 960
(B) 1040
(C) 1120
(D) 1200
(E) 1280

Q17: A container holds a mixture of juice and water in the ratio 5:3. If 8 litres of pure juice are added, the ratio of juice to water becomes 3:1. Find the difference between the initial quantities of juice and water.
एक बर्तन में जूस और पानी का मिश्रण 5:3 के अनुपात में है। यदि उसमें 8 लीटर शुद्ध जूस मिला दिया जाए, तो जूस और पानी का अनुपात 3:1 हो जाता है। प्रारंभ में जूस और पानी की मात्रा के बीच का अंतर ज्ञात कीजिए।
(A) 5
(B) 8
(C) 6
(D) 10
(E) 4

Q18: The total population of a town was 1,70,000. In the next few years, the male population increased by 20% and the female population increased by 8%, and the ratio of males to females then became 5:4. Find the original number of females.
एक नगर की कुल जनसंख्या 1,70,000 थी। अगले कुछ वर्षों में पुरुष जनसंख्या 20% बढ़ी और महिला जनसंख्या 8% बढ़ी, तथा उसके बाद पुरुषों और महिलाओं का अनुपात 5:4 हो गया। प्रारंभिक महिला जनसंख्या ज्ञात कीजिए।
(A) 90,000
(B) 80,000
(C) 85,000
(D) 75,000
(E) 70,000

Q19: Deepa invested Rs T for 2 years at 25% per annum compound interest, compounded annually. The amount obtained was then invested for 5 years at 16% per annum simple interest. If the simple interest earned in the second investment was Rs 20,000 more than T, find the value of T.
दीपा ने T रुपये 2 वर्षों के लिए 25% वार्षिक चक्रवृद्धि ब्याज पर, वार्षिक संयोजन के साथ निवेश किए। प्राप्त राशि को फिर 5 वर्षों के लिए 16% वार्षिक साधारण ब्याज पर निवेश किया गया। यदि दूसरे निवेश से प्राप्त साधारण ब्याज, T से 20,000 रुपये अधिक था, तो T का

मान ज्ञात कीजिए।
(A) 60,000
(B) 72,000
(C) 90,000
(D) 80,000
(E) 1,00,000

Q20: The total cost of painting the four walls and the ceiling of a room is Rs 6800. The painting rate is Rs 25 per square metre. If the length and height of the room are 14 metres and 6 metres respectively, find the breadth of the room.
एक कमरे की चारों दीवारों और छत को रंगने की कुल लागत 6800 रुपये है। रंगाई की दर 25 रुपये प्रति वर्ग मीटर है। यदि कमरे की लंबाई 14 मीटर और ऊँचाई 6 मीटर है, तो कमरे की चौड़ाई ज्ञात कीजिए।
(A) 4
(B) 5
(C) 6
(D) 3
(E) 8

Answers:

1. B
2. D
3. C
4. E
5. A
6. E
7. B
8. D
9. E
10. C
11. B
12. D
13. C
14. E
15. A
16. C
17. E
18. B
19. D
20. A

Solutions:

1. Option B

Let the marked prices of A and B be 7x and 9x respectively.
A discount of 20% is allowed on article A, so its selling price becomes 80% of its marked price.

Therefore, 80% of 7x = 560
So, 7x = 560 × 100 ÷ 80 = 700
Hence, x = 100
Therefore, marked price of article B = 9x = 9 × 100 = 900

2. Option D
Let the speeds of trains P and Q be 5x m/s and 7x m/s respectively.
Since both trains are moving in the same direction, relative speed = 7x − 5x = 2x
Total distance covered while crossing each other = sum of their lengths
= 220 + 308 = 528 m
Given time = 44 seconds
So, relative speed = 528 ÷ 44 = 12 m/s
Hence, 2x = 12
x = 6
Therefore, speed of train Q = 7x = 7 × 6 = 42 m/s

3. Option C

Take the total work as 48 units.
Y alone can complete the work in 24 days, so Y’s one-day work = 48 ÷ 24 = 2 units
Z is 50% more efficient than Y, so Z’s one-day work = 2 + 50% of 2 = 3 units
X, Y and Z together complete the work in 6 days, so their combined one-day work = 48 ÷ 6 = 8 units
Therefore, X’s one-day work = 8 − 2 − 3 = 3 units
Now X and Z together do in one day = 3 + 3 = 6 units
Hence, time taken by X and Z together = 48 ÷ 6 = 8 days

4. Option E

Total mixture = 90 litres
Milk : Water = 7 : 3
So, total parts = 10
Milk in the mixture = 90 × 7 ÷ 10 = 63 litres
Water in the mixture = 90 × 3 ÷ 10 = 27 litres
Now 15 litres of milk and 9 litres of water are added.
So, new milk quantity = 63 + 15 = 78 litres
New water quantity = 27 + 9 = 36 litres

5. Option A

Average weight of five boys = 44 kg
So, total weight of all five boys = 44 × 5 = 220 kg
Average weight of Anuj and Bharat = 42 kg
So, total weight of Anuj and Bharat = 42 × 2 = 84 kg
Average weight of Deepak and Eshan = 50 kg
So, total weight of Deepak and Eshan = 50 × 2 = 100 kg
Therefore, weight of Chetan = 220 − 84 − 100 = 36 kg

6. Option E

Since both Deepak and Sameer invested for the same period of time, their profit shares will be in the ratio of their investments.

Deepak's share = Rs 4500
Total profit = Rs 9900

So, Sameer's share = 9900 - 4500 = Rs 5400

Therefore, profit ratio of Deepak and Sameer
= 4500 : 5400
= 5 : 6

Now, investment ratio will also be 5 : 6

So,
x : (x + 900) = 5 : 6

Cross multiply:
6x = 5(x + 900)

6x = 5x + 4500

x = 4500

Hence, the value of x is Rs 4500.

7. Option B

Let the present ages of Harsh and Tarun be 5x and 3x respectively.

So, present sum of their ages
= 5x + 3x
= 8x

Eight years ago, each of them was 8 years younger.

Therefore, sum of their ages eight years ago
= 8x - 16

According to the question:
8x - 16 = 24

8x = 40

x = 5

So, Tarun's present age
= 3x
= 3 × 5
= 15 years

8. Option D

Let Megha's monthly income be Rs x.

She spends 25% on household needs.
So, remaining income
= 75% of x

Now she spends 20% of the remaining amount on coaching classes.
That means she keeps 80% of the remaining 75%.

So, amount left after coaching
= 80% of 75% of x
= 60% of x

Now she spends 30% of this remaining amount on transport.
That means she keeps 70% of 60%.

So, final savings
= 70% of 60% of x
= 42% of x

Given that savings = Rs 16800

Therefore,
42% of x = 16800

x = 16800 × 100 ÷ 42

x = 40000

Hence, Megha's monthly income is Rs 40000.

9. Option E

First calculate the simple interest earned from Scheme A.

Principal = Rs 12000
Rate = 15% per annum
Time = 4 years

Simple Interest
$= 12000 \times 15 \times 4 \div 100$
$= 7200$

So, the interest earned from Scheme A is Rs 7200.

Now this Rs 7200 is invested in Scheme B at 25% compound interest for 2 years.

Amount after 2 years
$= 7200 \times 125 \div 100 \times 125 \div 100$
$= 7200 \times 1.25 \times 1.25$
$= 11250$

Compound Interest
= Amount - Principal
$= 11250 - 7200$
$= 4050$

10. Option C

Given:
$A + B = 29$

So,
$A = 29 - B$

Also given:
$A^2 = 3B^2 + 61$

Now substitute $A = 29 - B$ into the second equation:

$(29 - B)^2 = 3B^2 + 61$

Now expand:

$29^2 - 2 \times 29 \times B + B^2 = 3B^2 + 61$

$841 - 58B + B^2 = 3B^2 + 61$

Bring all terms to one side:

$841 - 58B + B^2 - 3B^2 - 61 = 0$

$780 - 58B - 2B^2 = 0$

Multiply by -1:

$2B^2 + 58B - 780 = 0$

Divide the whole equation by 2:

$B^2 + 29B - 390 = 0$

Now factorise:

$(B + 39)(B - 10) = 0$

So,
$B = 10$
because B is a positive integer

Now,
$A = 29 - 10 = 19$

Therefore,
$A - B = 19 - 10 = 9$

11. Option B
Increase in selling price = Rs 180. This increase causes the profit percentage to rise by 15 percentage points. So,
15% of cost price = 180
Cost price = 180 × 100 ÷ 15 = 1200
Check:
At Rs 1500, profit = 1500 − 1200 = 300, so profit% = 25%
At Rs 1680, profit = 1680 − 1200 = 480, so profit% = 40%
Difference = 15 percentage points
Hence, the cost price is Rs 1200.

12. Option D
Let height = 4x and radius = 3x.
Volume of cone = π × r × r × h ÷ 3
So,
324π = π × 3x × 3x × 4x ÷ 3
324π = 12πx cube
324 = 12x cube
x cube = 27

x = 3
Therefore,
height = 12 cm and radius = 9 cm
Slant height = $\sqrt{12^2 + 9^2}$
= $\sqrt{144 + 81}$
= $\sqrt{225}$
= 15 cm

13. Option C

Let population in 2021 be P.
In 2022, population increased by 20%.
So,
120% of P = 7200
P = 7200 × 100 ÷ 120
P = 6000

14. Option E

Profit share of Rohit = Rs 2000
Profit share of Varun = 5000 − 2000 = Rs 3000
So profit ratio of Rohit and Varun = 2000 : 3000 = 2 : 3
Profit is proportional to investment × time.
Time ratio of Rohit and Varun = 3 : 2
So investment ratio of Rohit and Varun =
2 × 2 : 3 × 3
= 4 : 9
Total investment = Rs 7800
Total parts = 4 + 9 = 13
One part = 7800 ÷ 13 = 600
Varun's investment = 9 × 600 = Rs 5400

15. Option A
Average of 9 numbers = 46
So total sum of 9 numbers = 9 × 46 = 414
Average of first 4 numbers = 39
So sum of first 4 numbers = 4 × 39 = 156
Average of last 4 numbers = 50

So sum of last 4 numbers = 4 × 50 = 200
Therefore, 5th number = 414 − 156 − 200 = 58

16. Option C
In partnership, profit is shared in the ratio of capital × time.
Rohan invested 6000 for 5 months, so his money-months = 6000 × 5 = 30000.
Suresh invested 9000 for 12 months, so his money-months = 9000 × 12 = 108000.
Total initial investment of Rohan and Suresh = 6000 + 9000 = 15000.
Nitin joined with 80% of 15000, so Nitin's investment = 12000.
He stayed for the remaining 7 months, so his money-months = 12000 × 7 = 84000.
Now the profit ratio becomes:
30000 : 108000 : 84000
Divide by 6000:
5 : 18 : 14
Total parts = 5 + 18 + 14 = 37
Value of 1 part = 2960 ÷ 37 = 80
So Nitin's share = 14 × 80 = 1120

17. Option E

Let the initial quantities of juice and water be 5x and 3x respectively.
After adding 8 litres of juice, the quantities become:
Juice = 5x + 8
Water = 3x
According to the question:
5x + 8 : 3x = 3 : 1
So,
5x + 8 = 9x
4x = 8
x = 2
Therefore, initial juice = 5x = 10 litres
Initial water = 3x = 6 litres
Required difference = 10 − 6 = 4 litres

18. Option B

Let the original numbers of males and females be M and F.
After increase:
Male population = 120% of M

Female population = 108% of F
Given new ratio:
120% of M : 108% of F = 5 : 4
So,
M : F = 5 × 108 : 4 × 120
= 540 : 480
= 9 : 8
Thus, the original ratio of males to females was 9 : 8.
Total population = 1,70,000
Total parts = 9 + 8 = 17
Value of 1 part = 1,70,000 ÷ 17 = 10,000
Original number of females = 8 × 10,000 = 80,000

19. Option D

Deepa first invested T at 25% compound interest for 2 years.
Amount after 2 years:
= T × 125% × 125%
= T × 1.25 × 1.25
= 1.5625T
This amount was invested again for 5 years at 16% simple interest.
Simple interest on the second investment:
= 1.5625T × 16 × 5 ÷ 100
= 1.5625T × 0.8
= 1.25T
It is given that this simple interest was Rs 20,000 more than T.
So,
1.25T − T = 20,000
0.25T = 20,000
T = 20,000 ÷ 0.25
T = 80,000

20. Option A

Total cost of painting = Rs 6800
Rate of painting = Rs 25 per square metre
Therefore, total painted area = 6800 ÷ 25 = 272 square metres
Area painted includes four walls and the ceiling.
Area of four walls = 2h(l + b)

Area of ceiling = $l \times b$

Given:

$l = 14$ metres

$h = 6$ metres

Let breadth = b metres

So total area:

$2 \times 6 \times (14 + b) + 14b = 272$

$12(14 + b) + 14b = 272$

$168 + 12b + 14b = 272$

$168 + 26b = 272$

$26b = 104$

$b = 4$

3. Quadratic Equations

In each of the following questions, there are two equations. You have to solve both equations and mark the correct answer.

(a) $x > y$

(b) $x < y$

(c) $x = y$ or the relationship cannot be established

(d) $x \geq y$

(e) $x \leq y$

1.) I. $6x^2 - 41x + 65 = 0$

II. $2y^2 + 7y - 30 = 0$

2.) I. $6x^2 - 43x + 77 = 0$

II. $2y^2 - y - 15 = 0$

3.) I. $6x^2 - 29x + 35 = 0$

II. $2y^2 - 17y + 36 = 0$

4.) I. $3x^2 - 5x - 28 = 0$

II. $2y^2 - 19y + 44 = 0$

5.) I. $6x^2 - 41x + 70 = 0$

II. $6y^2 - 37y + 55 = 0$

6.) I. $x^2 - 3025 = 0$

II. $y - \sqrt[3]{166375} = 0$

7.) I. $20x^2 - 71x + 63 = 0$

II. $2y^2 - 11y + 15 = 0$

8.) I. $12x^2 - 91x + 170 = 0$

II. $3y^2 - 16y + 21 = 0$

9.) I. $3x^2 - 31x + 78 = 0$

II. $12y^2 - 85y + 143 = 0$

10.) I. $16x^2 - 80x + 99 = 0$

II. $3y^2 - 22y + 40 = 0$

11.) I. $8x^2 - 66x + 133 = 0$

II. $6y^2 - 35y + 50 = 0$

12.) I. $x^2 - 4489 = 0$

II. $(y + 67)^2 = 0$

13.) I. $6x^2 - 43x + 70 = 0$

II. $9y^2 - 57y + 88 = 0$

14.) I. $15x^2 - 68x + 77 = 0$

II. $6y^2 - 47y + 91 = 0$

15.) I. $20x^2 - 177x + 391 = 0$

II. $8y^2 - 54y + 91 = 0$

16.) I. $x^4 - 707281 = 0$

II. $y - (\sqrt{841} + \sqrt{4}) = 0$

17.) I. $9x^2 - 63x + 104 = 0$

II. $y^2 - 7y + 12 = 0$

18.) I. $8x^2 + 2x - 45 = 0$

II. $8y^2 - 46y + 63 = 0$

19.) I. $10x^2 - 67x + 112 = 0$

II. $20y^2 - 119y + 176 = 0$

20.) I. $12x^2 - 125x + 323 = 0$

II. $6y^2 - 53y + 117 = 0$

Answers:

1. D

2. A

3. B

4. E

5. C

6. E

7. B

8. A

9. D

10. B

11. A

12. D

13. C

14. B

15. A

16. B

17. C

18. E

19. D

20. A

Solutions:

(1) x = $\frac{13}{3}$, $\frac{5}{2}$; y = $\frac{5}{2}$, -6

(2) x = $\frac{11}{3}$, $\frac{7}{2}$; y = 3, -$\frac{5}{2}$

(3) x = $\frac{5}{2}$, $\frac{7}{3}$; y = $\frac{9}{2}$, 4

(4) x = 4, -$\frac{7}{3}$; y = $\frac{11}{2}$, 4

(5) x = $\frac{7}{2}$, $\frac{10}{3}$; y = $\frac{11}{3}$, $\frac{5}{2}$

(6) x = 55, -55 ; y = 55

(7) x = $\frac{9}{5}$, $\frac{7}{4}$; y = 3, $\frac{5}{2}$

(8) x = $\frac{17}{4}$, $\frac{10}{3}$; y = 3, $\frac{7}{3}$

(9) x = 6, $\frac{13}{3}$; y = $\frac{13}{3}$, $\frac{11}{4}$

(10) x = $\frac{11}{4}$, $\frac{9}{4}$; y = 4, $\frac{10}{3}$

(11) x = $\frac{19}{4}$, $\frac{7}{2}$; y = $\frac{10}{3}$, $\frac{5}{2}$

(12) x = 67, -67 ; y = -67

(13) x = $\frac{14}{3}$, $\frac{5}{2}$; y = $\frac{11}{3}$, $\frac{8}{3}$

(14) x = $\frac{11}{5}$, $\frac{7}{3}$; y = $\frac{13}{3}$, $\frac{7}{2}$

(15) x = $\frac{23}{5}$, $\frac{17}{4}$; y = $\frac{7}{2}$, $\frac{13}{4}$

(16) x = 29, -29 ; y = 31

(17) x = $\frac{13}{3}$, $\frac{8}{3}$; y = 4, 3

(18) x = $\frac{9}{4}$, -$\frac{5}{2}$; y = $\frac{7}{2}$, $\frac{9}{4}$

(19) x = $\frac{7}{2}$, $\frac{16}{5}$; y = $\frac{16}{5}$, $\frac{11}{4}$

(20) x = $\frac{17}{3}$, $\frac{19}{4}$; y = $\frac{9}{2}$, $\frac{13}{3}$

4. WRONG NUMBER SERIES

(1) 1836, 613, 306, 102, 51, 17

(a) 613

(b) 1836

(c) 51

(d) 306

(e) None of these

(2) 285, 294.5, 305, 310.5, 317, 322.5

(a) 285

(b) 305

(c) 317

(d) 322.5

(e) None of these

(3) 215, 38, 253, 291, 544, 838

(a) 215

(b) 253

(c) 291

(d) 838

(e) None of these

(4) 625, 619, 612, 600, 581, 553

(a) 553

(b) 581

(c) 612

(d) 625

(e) None of these

(5) 166, 156.5, 150, 140.5, 134, 128.5

(a) 134

(b) 166

(c) 150

(d) 128.5

(e) None of these

(6) 58, 77, 106, 154, 218, 298

(a) 58

(b) 77

(c) 154

(d) 298

(e) None of these

(7) 8, 35, 144, 581, 2330, 9329

(a) 9329

(b) 8

(c) 144

(d) 581

(e) None of these

(8) 10, 28, 110, 541, 3241, 22681

(a) 3241

(b) 28

(c) 10

(d) 110

(e) None of these

(9) 6, 17, 45, 129, 376, 1110

(a) 1110

(b) 376

(c) 45

(d) 17

(e) None of these

(10) 37, 49, 73, 112, 157, 217

(a) 49

(b) 37

(c) 112

(d) 217

(e) None of these

(11) 144, 155, 164, 180, 200, 224

(a) 155

(b) 144

(c) 180

(d) 224

(e) None of these

(12) 38, 159, 259, 340, 404, 454

(a) 38

(b) 454

(c) 259

(d) 404

(e) None of these

(13) 172, 200.5, 238.5, 288, 343, 409.5

(a) 238.5

(b) 409.5

(c) 172

(d) 288

(e) None of these

(14) 45, 63, 86, 107, 137, 175

(a) 86

(b) 45

(c) 107

(d) 175

(e) None of these

(15) 23, 45, 68, 95, 125, 158

(a) 95

(b) 23

(c) 45

(d) 158

(e) None of these

(16) 280, 39, 319, 360, 677, 1035

(a) 360

(b) 280

(c) 319

(d) 677

(e) None of these

(17) 14, 181, 325, 446, 546, 627

(a) 446

(b) 181

(c) 627

(d) 14

(e) None of these

(18) 8192, 1024, 512, 64, 32, 2

(a) 512

(b) 2

(c) 32

(d) 8192

(e) None of these

(19) 89, 85, 161, 467, 1845, 9179

(a) 1845

(b) 89

(c) 161

(d) 467

(e) None of these

(20) 815, 791, 759, 720, 671, 615

(a) 815

(b) 720

(c) 615

(d) 671

(e) None of these

Answers:

(1) a

(2) b

(3) d

(4) d

(5) c

(6) b

(7) a

(8) d

(9) d

(10) c

(11) a

(12) b

(13) d

(14) a

(15) c

(16) a

(17) d

(18) b

(19) a

(20) b

Solutions:

(1) ÷3, ÷2, ÷3, ÷2, ÷3

(2) +9.5, +8.5, +7.5, +6.5, +5.5

(3) Sum of the previous two numbers

(4) -4 -7 -12 -19 -28
+3 +5 +7 +9

(5) -9.5, -8.5, -7.5, -6.5, -5.5

(6) +16, +32, +48, +64, +80

(7) x4+3, x4+4, x4+5, x4+6, x4+7

(8) x3-2, x4-3, x5-4, x6-5, x7-6

(9) $+3^2+1$, $+3^3+2$, $+3^4+3$, $+3^5+4$, $+3^6+5$

(10) +12, +24, +36, +48, +60

(11) +4x2, +4x3, +4x4, +4x5, +4x6

(12) $+11^2$, $+10^2$, $+9^2$, $+8^2$, $+7^2$

(13) +3x9.5, +4x9.5, +5x9.5, +6x9.5, +7x9.5

(14) +18 +20 +24 +30 +38
+2 +4 +6 +8

(15) +3x7, +3x8, +3x9, +3x10, +3x11

(16) Sum of the previous two numbers

(17) $+13^2$, $+12^2$, $+11^2$, $+10^2$, $+9^2$

(18) ÷8, ÷2, ÷8, ÷2, ÷8

(19) $x1-2^2$, $x2-3^2$, $x3-4^2$, $x4-5^2$, $x5-6^2$

(20) -3x8, -4x8, -5x8, -6x8, -7x8

5. MISSING NUMBER SERIES

(1) 124, 129, ?, 159, 188, 229
(a) 136
(b) 140
(c) 142
(d) 144
(e) None of these

(2) 156, 310, 616, ?, 2432, 4832
(a) 1218
(b) 1220
(c) 1222
(d) 1224
(e) None of these

(3) 567, 557, ?, 521, 493, 457
(a) 542
(b) 540
(c) 544
(d) 546
(e) None of these

(4) 125.5, 130, 137.5, 148, ?, 178
(a) 157.5
(b) 159.5
(c) 161.5
(d) 163.5
(e) None of these

(5) 248, 233, 230, 260, ?, 296
(a) 204
(b) 206
(c) 208
(d) 210
(e) None of these

(6) 324, 313, 300, ?, 264, 241
(a) 277
(b) 279
(c) 281
(d) 283
(e) None of these

(7) 542, 554, 590, ?, 734, 842
(a) 650
(b) 648
(c) 652
(d) 654
(e) None of these

(8) 187, 203, 233, ?, 351, 447
(a) 277

(b) 279
(c) 281
(d) 283
(e) None of these

(9) 620, 637, 624, ?, 628, 657
(a) 645
(b) 647
(c) 649
(d) 651
(e) None of these

(10) 150, 154, 308, ?, 634, 650
(a) 311
(b) 313
(c) 315
(d) 317
(e) None of these

(11) 1970, ?, 340, 95, 30, 17
(a) 993
(b) 991
(c) 995
(d) 997
(e) None of these

(12) 124, 148, ?, 222, 276, 344
(a) 176
(b) 178
(c) 180
(d) 182
(e) None of these

(13) 210, 627, 1875, ?, 16830, 50469
(a) 5605
(b) 5615
(c) 5625
(d) 5635
(e) None of these

(14) 1025, 998, 944, ?, 755, 620
(a) 857
(b) 859
(c) 861
(d) 863
(e) None of these

(15) 375.5, 380, 388.5, 401, ?, 438
(a) 417.5
(b) 419.5
(c) 421.5
(d) 423.5
(e) None of these

(16) 432, 439, 425, ?, 418, 453
(a) 442
(b) 444
(c) 446
(d) 448
(e) None of these

(17) 189, 201, 223, ?, 313, 389
(a) 257
(b) 259
(c) 261
(d) 263
(e) None of these

(18) 208, 253, 314, ?, 492, 613
(a) 393
(b) 395
(c) 397

(d) 399
(e) None of these

(19) 420, 404, 379, ?, 294, 230
(a) 339
(b) 341
(c) 343
(d) 345
(e) None of these

(20) 250, 263, 515, 532, ?, 1068
(a) 1039
(b) 1041
(c) 1043
(d) 1045
(e) None of these

Answers:

(1) b
(2) d
(3) a
(4) c
(5) b
(6) d
(7) a
(8) c
(9) b
(10) d
(11) a
(12) c
(13) b
(14) d
(15) a
(16) c
(17) b
(18) a
(19) c
(20) d

Solutions:

(1) +5, +11, +19, +29, +41

+6 +8 +10 +12

(2) ×2 − 2, ×2 − 4, ×2 − 8, ×2 − 16, ×2 − 32

(3) −10, −15, −21, −28, −36

(4) +4.5, +7.5, +10.5, +13.5, +16.5

(5) Odd positions: −18, −24
Even positions: +27, +36

(6) −11, −13, −17, −19, −23

(7) +12x1, +12x3, +12x5, +12x7, +12x9

(8) +16, +30, +48, +70, +96

+14 +18 +22 +26

(9) +17, −13, +23, −19, +29

(10) +4, ×2, +9, ×2, +16

(11) ÷2 + 8, ÷3 + 9, ÷4 + 10, ÷5 + 11, ÷6 + 12

(12) +24, +32, +42, +54, +68

+8 +10 +12 +14

(13) ×3 − 3, ×3 − 6, ×3 − 10, ×3 − 15, ×3 − 21

(14) −27, −54, −81, −108, −135

(15) +4.5, +8.5, +12.5, +16.5, +20.5

(16) +7, −14, +21, −28, +35

(17) +12, +22, +36, +54, +76

+10 +14 +18 +22

(18) +45, +61, +79, +99, +121

+16 +18 +20 +22

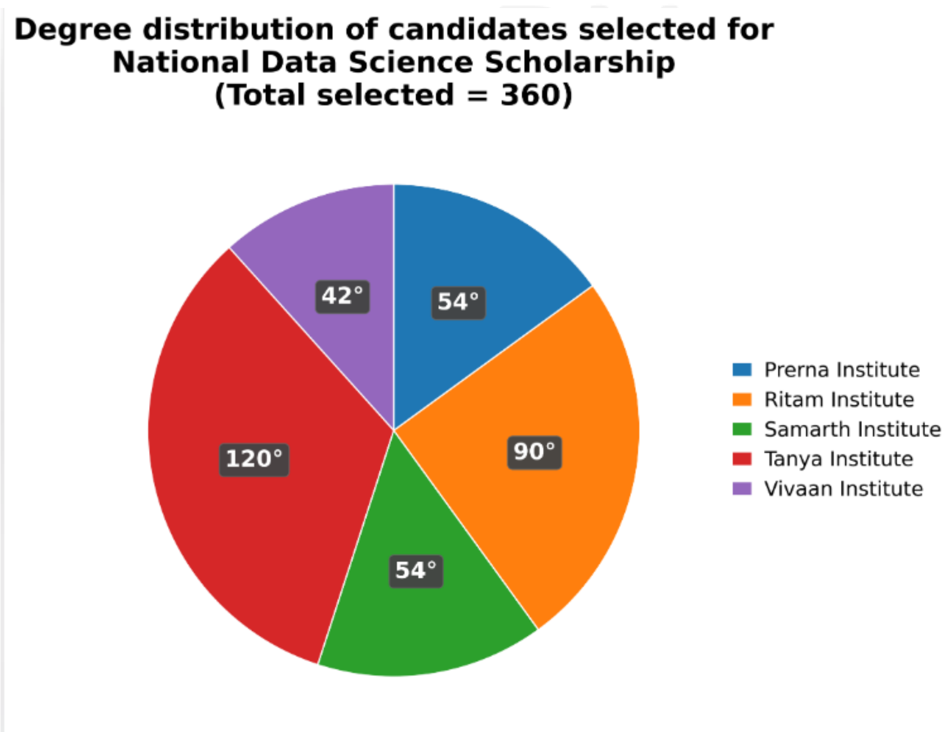
(19) $-4^2, -5^2, -6^2, -7^2, -8^2$

(20) +13, ×2 − 11, +17, ×2 − 19, +23

6. DATA INTERPRETATION

Set 1: The pie chart shows the degree-wise distribution of candidates selected for a National Data Science Scholarship from five different coaching institutes. Read the chart carefully and answer the following questions.

पाई चार्ट पाँच अलग-अलग कोचिंग संस्थानों से नेशनल डेटा साइंस स्कॉलरशिप के लिए चयनित अभ्यर्थियों का डिग्री-वार वितरण दर्शाता है। चार्ट को ध्यानपूर्वक पढ़िए और निम्नलिखित प्रश्नों के उत्तर दीजिए।



Q1: If the ratio of boys to girls who cleared the scholarship test from Samarth Institute is 2:7, and the number of girls in Ritam Institute who qualified is 12 more than that of the girls in Samarth Institute, then find the number of boys who cleared the scholarship test from Ritam Institute.

यदि समर्थ संस्थान से स्कॉलरशिप टेस्ट उत्तीर्ण करने वाले लड़कों और लड़कियों का अनुपात 2:7 है, और ऋतम संस्थान में चयनित लड़कियों की संख्या, समर्थ संस्थान की लड़कियों से 12

अधिक है, तो ऋतम संस्थान से स्कॉलरशिप टेस्ट उत्तीर्ण करने वाले लड़कों की संख्या ज्ञात कीजिए।
(A) 24
(B) 30
(C) 42
(D) 48
(E) 36

Q2: The ratio of boys to girls who cleared the scholarship test from Ritam Institute is 2:3. In Yashvi Institute, the number of boys who qualified is 18 more than that of Ritam Institute, and the number of girls who qualified is 6 less than that of Ritam Institute. Find the total number of students who cleared the scholarship test from Yashvi Institute.
ऋतम संस्थान से स्कॉलरशिप टेस्ट उत्तीर्ण करने वाले लड़कों और लड़कियों का अनुपात 2:3 है। यश्वी संस्थान में चयनित लड़कों की संख्या ऋतम संस्थान से 18 अधिक है तथा चयनित लड़कियों की संख्या ऋतम संस्थान से 6 कम है। यश्वी संस्थान से स्कॉलरशिप टेस्ट उत्तीर्ण करने वाले कुल छात्रों की संख्या ज्ञात कीजिए।
(A) 96
(B) 102
(C) 108
(D) 114
(E) 120

Q3: From Tanya Institute, 75% students did not clear the scholarship test. Find the total number of students in Tanya Institute.
तान्या संस्थान से 75% छात्र स्कॉलरशिप टेस्ट पास नहीं कर सके। तान्या संस्थान में छात्रों की कुल संख्या ज्ञात कीजिए।
(A) 420
(B) 450
(C) 510
(D) 480
(E) 540

Q4: Students who cleared the scholarship test from Ritam Institute are what percentage more or less than the students who cleared it from Tanya Institute?
ऋतम संस्थान से स्कॉलरशिप टेस्ट उत्तीर्ण करने वाले छात्रों की संख्या, तान्या संस्थान से स्कॉलरशिप टेस्ट उत्तीर्ण करने वाले छात्रों की संख्या की तुलना में कितने प्रतिशत अधिक या

कम है?
(A) 25% less
(B) 20% more
(C) 15% less
(D) 25% more
(E) 20% less

Q5: If the ratio of students who cleared the scholarship test from Prerna Institute to Tanya Institute is p:q in the simplest form, then find the value of p+q.
यदि प्रेरणा संस्थान और तान्या संस्थान से स्कॉलरशिप टेस्ट उत्तीर्ण करने वाले छात्रों का अनुपात सरलतम रूप में p:q है, तो p+q का मान ज्ञात कीजिए।
(A) 25
(B) 27
(C) 29
(D) 31
(E) 33

Solutions:

1. Option E
Selected students in Samarth Institute = 54
Boys : Girls = 2 : 7
Total parts = 9
1 part = 54/9 = 6
Girls in Samarth = 7 × 6 = 42
Girls in Ritam = 42 + 12 = 54
Selected students in Ritam = 90
Boys in Ritam = 90 − 54 = 36

2. Option B
Selected students in Ritam = 90
Boys : Girls = 2 : 3
Total parts = 5
1 part = 90/5 = 18
Boys in Ritam = 2 × 18 = 36
Girls in Ritam = 3 × 18 = 54
Boys in Yashvi = 36 + 18 = 54

Girls in Yashvi = 54 − 6 = 48
Total students in Yashvi = 54 + 48 = 102

3. Option D
Selected students in Tanya = 120
75% did not clear, so 25% cleared
25% of total students = 120
Total students = 120 × 100/25 = 480

4. Option A
Selected students in Ritam = 90
Selected students in Tanya = 120
Difference = 120 − 90 = 30
Required percentage = 30/120 × 100 = 25
So, Ritam is 25% less than Tanya

5. Option C
Selected students in Prerna = 54
Selected students in Tanya = 120
Required ratio = 54 : 120 = 9 : 20
p + q = 9 + 20 = 29

Set 2: The table chart shows the total number of households in five Indian districts and the percentage of households using rooftop solar systems. Read the data carefully and answer the following questions.
तालिका चार्ट पाँच भारतीय जिलों में कुल परिवारों की संख्या तथा rooftop solar system उपयोग करने वाले परिवारों का प्रतिशत दिखाता है। दिए गए आँकड़ों को ध्यानपूर्वक पढ़िए और निम्नलिखित प्रश्नों के उत्तर दीजिए।
Note: Total households = Solar households + Non-solar households.
नोट: कुल परिवार = Solar परिवार + Non-solar परिवार।

District	Total households (in thousands)	Percentage of households using rooftop solar
Ajmer	192	62.5%
Bhopal	384	75%
Cuttack	336	37.5%
Dharwad	480	58(1/3)%
Erode	576	66(2/3)%

Q1: If the simplest ratio of the number of non-solar households in Cuttack to the number of non-solar households in Erode is a : b, then find the value of a + b.
यदि कटक के non-solar परिवारों की संख्या और ईरोड के non-solar परिवारों की संख्या का सरलतम अनुपात a : b है, तो a + b का मान ज्ञात कीजिए।
(A) 61
(B) 63
(C) 65
(D) 67
(E) 69

Q2: The number of solar households in Bhopal is what percent of the number of non-solar households in Ajmer?
भोपाल के solar परिवारों की संख्या, अजमेर के non-solar परिवारों की संख्या का कितने प्रतिशत है?
(A) 400
(B) 350
(C) 375
(D) 425
(E) 450

Q3: Find the difference between the number of solar households in Erode and the number of solar households in Bhopal.
ईरोड के solar परिवारों की संख्या और भोपाल के solar परिवारों की संख्या के बीच अंतर ज्ञात कीजिए।
(A) 88
(B) 92
(C) 104
(D) 84
(E) 96

Q4: If the total number of households in Faridkot is 75% of the total households in Erode and the number of non-solar households in Faridkot is 1/5 of the total non-solar households in Cuttack and Dharwad together, then find the number of solar households in Faridkot.
यदि फरीदकोट में कुल परिवारों की संख्या, ईरोड के कुल परिवारों की 75% है और फरीदकोट के non-solar परिवारों की संख्या, कटक और धारवाड़ के कुल non-solar परिवारों की 1/5 है, तो फरीदकोट में solar परिवारों की संख्या ज्ञात कीजिए।
(A) 330
(B) 340
(C) 350
(D) 360
(E) 370

Q5: Find the average number of solar households in Ajmer, Bhopal and Cuttack.
अजमेर, भोपाल और कटक में solar परिवारों की औसत संख्या ज्ञात कीजिए।
(A) 172
(B) 178
(C) 184
(D) 176
(E) 180

Solutions:

Solved solar and non solar household counts with final answers

District	Total households	Solar households	Non-solar households
Ajmer	192	120	72
Bhopal	384	288	96
Cuttack	336	126	210
Dharwad	480	280	200
Erode	576	384	192

1. Option D
Non-solar households in Cuttack = 336 - 3/8 of 336 = 336 - 126 = 210
Non-solar households in Erode = 576 - 2/3 of 576 = 576 - 384 = 192
Ratio = 210 : 192 = 35 : 32
a + b = 67

2. Option A
Solar households in Bhopal = 3/4 of 384 = 288
Non-solar households in Ajmer = 192 - 5/8 of 192 = 72
Required percent = 288/72 × 100 = 400

3. Option E
Solar households in Erode = 2/3 of 576 = 384
Solar households in Bhopal = 3/4 of 384 = 288
Difference = 384 - 288 = 96

4. Option C
Total households in Faridkot = 3/4 of 576 = 432
Non-solar households in Cuttack = 210
Non-solar households in Dharwad = 480 - 7/12 of 480 = 200
Non-solar households in Faridkot = 1/5 of (210 + 200) = 82
Solar households in Faridkot = 432 - 82 = 350

5. Option B
Solar households in Ajmer = 5/8 of 192 = 120
Solar households in Bhopal = 288
Solar households in Cuttack = 126
Average = (120 + 288 + 126) / 3 = 534 / 3 = 178

Set 3: The table shows the number of ceramic mugs produced by three pottery studios—Asha, Bharat and Chitra—in four different years. Some data in the table is missing. Wherever a percentage is given in the table, it represents that percentage of the total mugs produced in that year. Read the table carefully and answer the following questions.

तालिका चार अलग-अलग वर्षों में तीन पॉटरी स्टूडियो—आशा, भारत और चित्रा—द्वारा बनाए गए सिरेमिक मगों की संख्या को दर्शाती है। तालिका के कुछ आँकड़े लुप्त हैं। तालिका में जहाँ भी प्रतिशत दिया गया है, वह उस वर्ष के कुल बने मगों में उस स्टूडियो की हिस्सेदारी को दर्शाता है। तालिका को ध्यानपूर्वक पढ़िए और निम्नलिखित प्रश्नों के उत्तर दीजिए।

Year	Asha Pottery Studio	Bharat Pottery Studio	Chitra Pottery Studio	Total
2024	1344	--	20%	4480
2025	1232	20%	--	3520
2026	--	20%	1680	5040
2027	1792	1344	1456	--

Q1: The simplified ratio between the total number of ceramic mugs produced by Bharat in 2026 and the total number of ceramic mugs produced by Asha in 2027 is p : q. Find the value of (p + q).
2026 में भारत स्टूडियो द्वारा बनाए गए सिरेमिक मगों की संख्या और 2027 में आशा स्टूडियो द्वारा बनाए गए सिरेमिक मगों की संख्या का सरल अनुपात p : q है। (p + q) का मान ज्ञात कीजिए।
(A) 25
(B) 21
(C) 27
(D) 29
(E) 31

Q2: Find the total number of ceramic mugs produced by Chitra in all four years together.
चित्रा स्टूडियो द्वारा चारों वर्षों में मिलाकर बनाए गए सिरेमिक मगों की कुल संख्या ज्ञात कीजिए।
(A) 5488
(B) 5568
(C) 5696
(D) 5616

(E) 5728

Q3: The total number of ceramic mugs produced by Chitra in 2025 is what percent more than the total number of ceramic mugs produced by Bharat in 2025?
2025 में चित्रा स्टूडियो द्वारा बनाए गए सिरेमिक मग, 2025 में भारत स्टूडियो द्वारा बनाए गए सिरेमिक मगों से कितने प्रतिशत अधिक हैं?
(A) 140
(B) 125
(C) 150
(D) 160
(E) 175

Q4: In 2026, Bharat Studio produced 216 damaged mugs. The ratio of damaged mugs produced by Bharat to Asha in 2026 is 9 : 7. Find the number of non-damaged mugs produced by Asha in 2026.
2026 में भारत स्टूडियो ने 216 क्षतिग्रस्त मग बनाए। 2026 में भारत और आशा स्टूडियो द्वारा बनाए गए क्षतिग्रस्त मगों का अनुपात 9 : 7 है। 2026 में आशा स्टूडियो द्वारा बनाए गए गैर-क्षतिग्रस्त मगों की संख्या ज्ञात कीजिए।
(A) 2104
(B) 2148
(C) 2200
(D) 2240
(E) 2184

Q5: Find the average number of ceramic mugs produced by all the three studios in 2026.
2026 में तीनों स्टूडियो द्वारा बनाए गए सिरेमिक मगों की औसत संख्या ज्ञात कीजिए।
(A) 1600
(B) 1640
(C) 1680
(D) 1720
(E) 1760

Solutions:

Year	Asha	Bharat	Chitra	Total
2024	1344	2240	896	4480
2025	1232	704	1584	3520
2026	2352	1008	1680	5040
2027	1792	1344	1456	4592

1. Option A
Bharat in 2026 = 20% of 5040 = 1008
Asha in 2027 = 1792
Required ratio = 1008 : 1792 = 9 : 16
p + q = 9 + 16 = 25

2. Option D
Chitra in 2024 = 20% of 4480 = 896
Bharat in 2025 = 20% of 3520 = 704
Chitra in 2025 = 3520 - 1232 - 704 = 1584
Total Chitra mugs in all four years = 896 + 1584 + 1680 + 1456 = 5616

3. Option B
Bharat in 2025 = 20% of 3520 = 704
Chitra in 2025 = 3520 - 1232 - 704 = 1584
Required percentage = (1584 - 704) / 704 × 100
= 880 / 704 × 100
= 125

4. Option E
Bharat damaged mugs in 2026 = 216
Damaged Bharat : Damaged Asha = 9 : 7
Damaged Asha = 216 × 7 / 9 = 168
Asha in 2026 = 5040 - 1008 - 1680 = 2352
Non-damaged Asha mugs = 2352 - 168 = 2184

5. Option C
Asha in 2026 = 5040 - 1008 - 1680 = 2352
Bharat in 2026 = 1008
Chitra in 2026 = 1680
Average in 2026 = (2352 + 1008 + 1680) / 3

= 5040 / 3
= 1680

Set 4: The data is about three art students, Aarohi, Bharat and Charvi, who went to a wholesale market to purchase canvas boards, paintbrush sets and colour palettes. Charvi purchased canvas boards, paintbrush sets and colour palettes in the ratio 2 : 3 : 5. The number of canvas boards purchased by Bharat was 50% more than the number of canvas boards purchased by Charvi. The ratio of canvas boards purchased by Aarohi to Bharat was 7 : 6. The total number of canvas boards purchased by Aarohi, Bharat and Charvi was 272. The ratio of paintbrush sets purchased by Aarohi to Bharat was 7 : 9. The number of colour palettes purchased by Aarohi was 50% more than the number of colour palettes purchased by Charvi. Bharat purchased 288 colour palettes. The sum of all items (canvas boards, paintbrush sets and colour palettes) purchased by Aarohi was 520.

दिया गया डेटा तीन कला विद्यार्थियों, आरोही, भारत और चार्वी, के बारे में है जो थोक बाजार से कैनवास बोर्ड, पेंटब्रश सेट और कलर पैलेट खरीदने गए। चार्वी ने कैनवास बोर्ड, पेंटब्रश सेट और कलर पैलेट 2 : 3 : 5 के अनुपात में खरीदे। भारत द्वारा खरीदे गए कैनवास बोर्ड, चार्वी द्वारा खरीदे गए कैनवास बोर्ड से 50% अधिक थे। आरोही और भारत द्वारा खरीदे गए कैनवास बोर्ड का अनुपात 7 : 6 था। आरोही, भारत और चार्वी द्वारा खरीदे गए कुल कैनवास बोर्ड 272 थे। आरोही और भारत द्वारा खरीदे गए पेंटब्रश सेट का अनुपात 7 : 9 था। आरोही द्वारा खरीदे गए कलर पैलेट, चार्वी द्वारा खरीदे गए कलर पैलेट से 50% अधिक थे। भारत ने 288 कलर पैलेट खरीदे। आरोही द्वारा खरीदी गई सभी वस्तुओं (कैनवास बोर्ड, पेंटब्रश सेट और कलर पैलेट) का कुल योग 520 था।

Q1: If the simplified ratio of the number of canvas boards purchased by Bharat to the number of colour palettes purchased by Bharat is x : y, find the value of x + y.
यदि भारत द्वारा खरीदे गए कैनवास बोर्ड और भारत द्वारा खरीदे गए कलर पैलेट का सरलतम अनुपात x : y है, तो x + y का मान ज्ञात कीजिए।
(A) 2
(B) 3
(C) 5
(D) 4
(E) 6

Q2: Find the sum of all items (canvas boards, paintbrush sets and colour palettes) purchased by Aarohi, Bharat and Charvi together.
आरोही, भारत और चार्वी द्वारा मिलाकर खरीदी गई सभी वस्तुओं (कैनवास बोर्ड, पेंटब्रश सेट और कलर पैलेट) का कुल योग ज्ञात कीजिए।
(A) 1408

(B) 1440
(C) 1472
(D) 1504
(E) 1536

Q3: The paintbrush sets purchased by Aarohi are what percent more than the colour palettes purchased by Charvi?
आरोही द्वारा खरीदे गए पेंटब्रश सेट, चार्वी द्वारा खरीदे गए कलर पैलेट से कितने प्रतिशत अधिक हैं?
(A) 10
(B) 8
(C) 6
(D) 12
(E) 5

Q4: Find the average number of paintbrush sets purchased by all the three persons.
तीनों व्यक्तियों द्वारा खरीदे गए पेंटब्रश सेटों की औसत संख्या ज्ञात कीजिए।
(A) 160
(B) 168
(C) 152
(D) 176
(E) 184

Q5: If the cost price of each paintbrush set is Rs. 25, then find the total expenditure of Aarohi on paintbrush sets.
यदि प्रत्येक पेंटब्रश सेट का क्रय मूल्य 25 रुपये है, तो पेंटब्रश सेटों पर आरोही का कुल व्यय ज्ञात कीजिए।
(A) 4000
(B) 3600
(C) 4200
(D) 4400
(E) 4600

Solutions:

Person	Canvas boards	Paintbrush sets	Colour palettes	Total
Aarohi	112	168	240	**520**
Bharat	96	216	288	**600**
Charvi	64	96	160	**320**
Total	**272**	**480**	**688**	**1440**

1. Option D
Required ratio = 96 : 288
= 1 : 3
x + y = 1 + 3 = 4

2. Option B
Total items purchased by all three = 520 + 600 + 320
= 1440

3. Option E
Required percent more = (168 − 160) / 160 × 100
= 8 / 160 × 100
= 5

4. Option A
Average paintbrush sets = (168 + 216 + 96) / 3
= 480 / 3
= 160

5. Option C
Total expenditure of Aarohi on paintbrush sets = 168 × 25
= 4200